

CHAPTER 20

Blood Vessels and Circulation

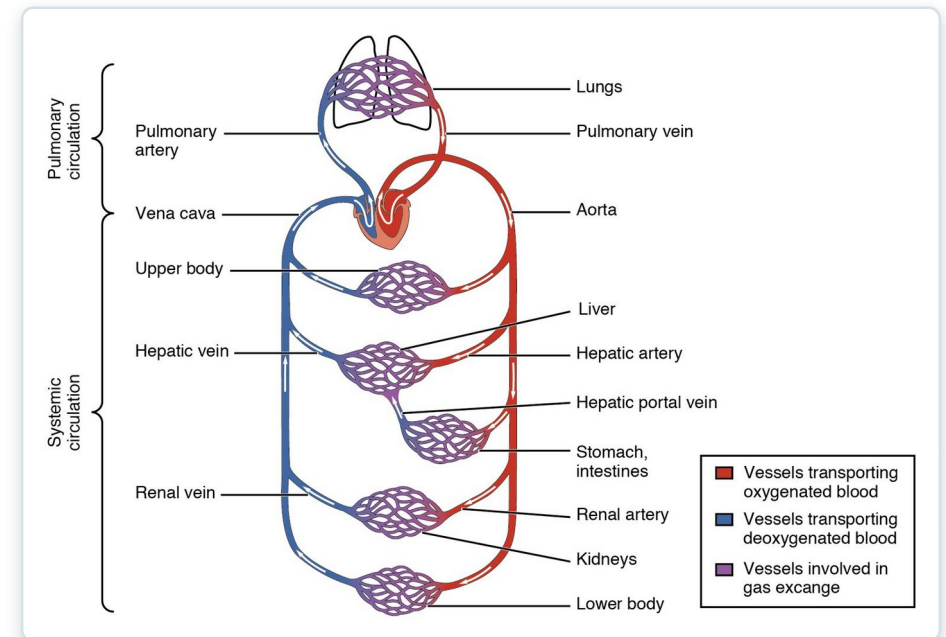
Vessels, blood pressure, exchange, and the circulatory pathways.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Compare and contrast the anatomical structure of arteries, arterioles, capillaries, venules, and veins
- 2 Accurately describe the forces that account for capillary exchange
- 3 List the major factors affecting blood flow, blood pressure, and resistance
- 4 Describe how blood flow, blood pressure, and resistance interrelate
- 5 Discuss how the neural and endocrine mechanisms maintain homeostasis within the blood vessels
- 6 Describe the interaction of the cardiovascular system with other body systems



The pulmonary and systemic circuits.

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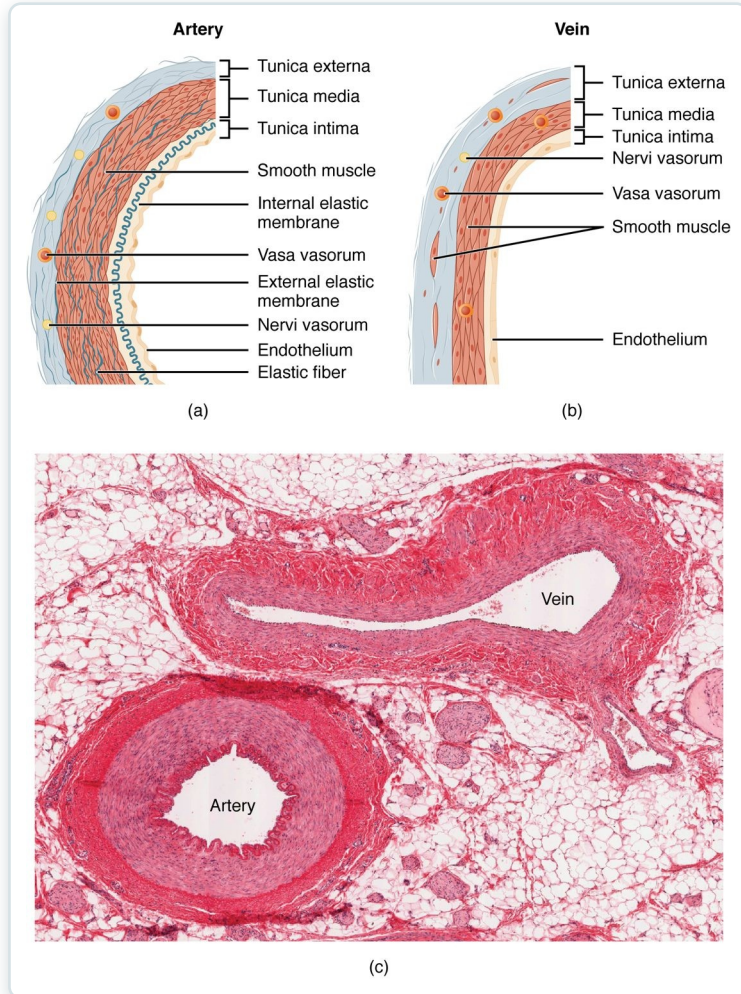
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SECTION

Structure and Function of Blood Vessels

The structure of arteries, veins, and capillaries.

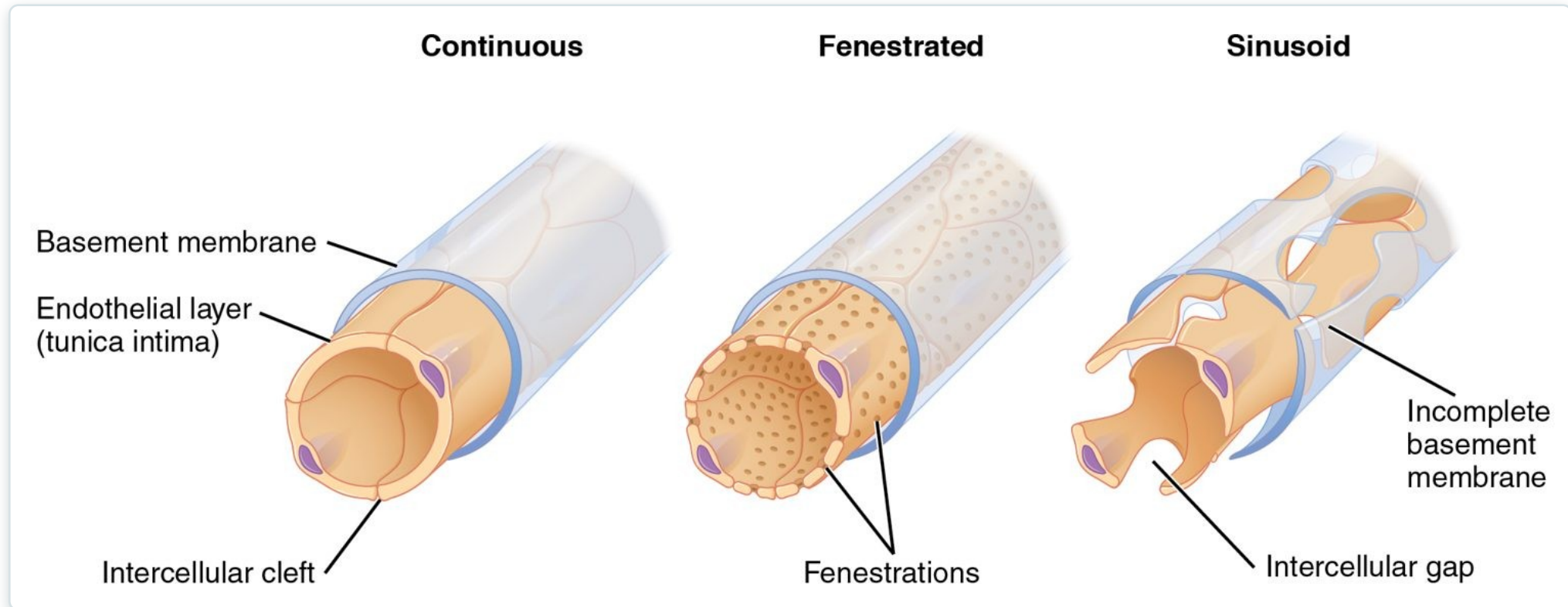
Artery vs. Vein



The walls of an artery and a vein

- Both have three wall layers (tunics)
- Arteries have thick muscular walls
- Veins are thinner, with valves
- Valves keep venous blood moving up

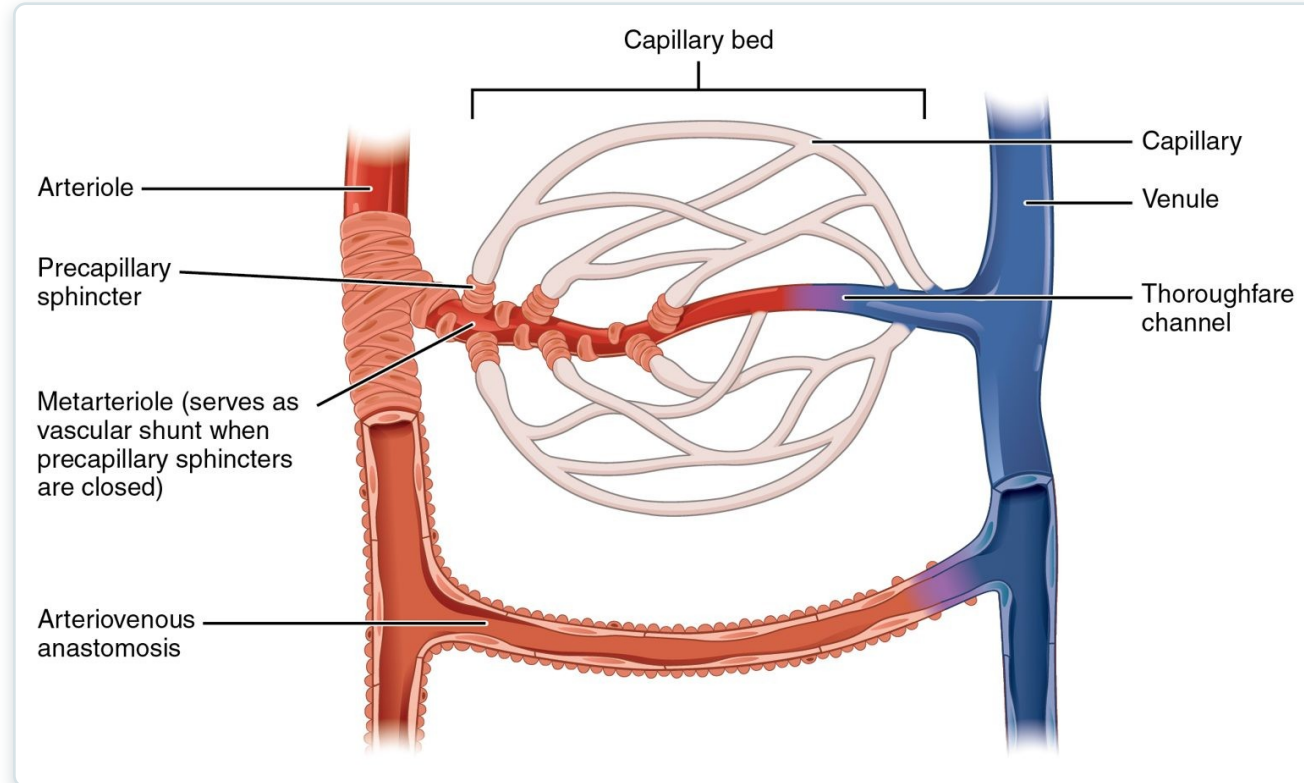
Types of Capillaries



The three types of capillaries

- The smallest vessels
- Walls just one cell thick
- Continuous, fenestrated, sinusoid
- The site of all exchange

The Capillary Bed



Blood flow through a capillary bed

- Capillaries form branching beds
- Connect arterioles to venules
- Precapillary sphincters control flow
- Direct blood where it's needed

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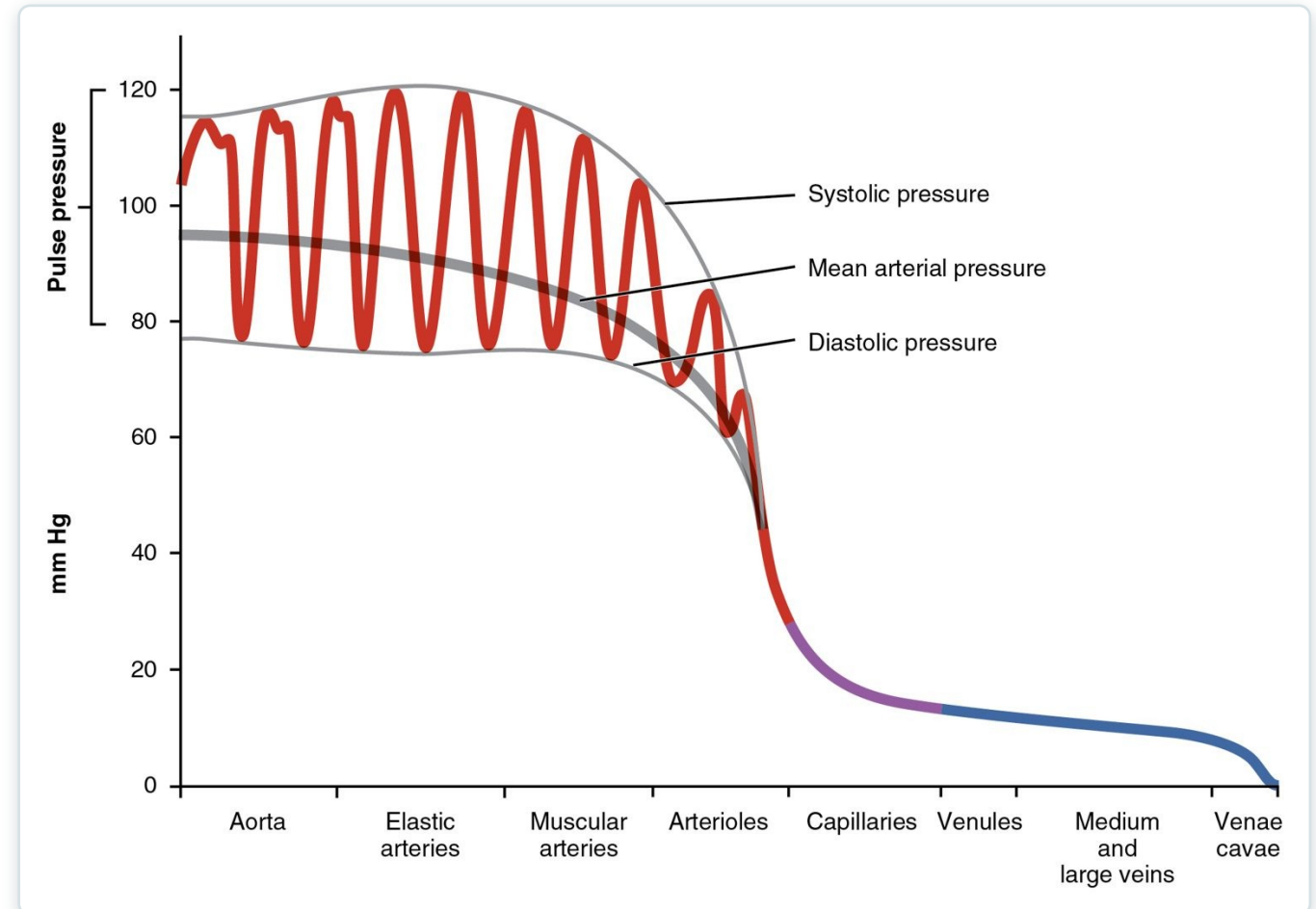
SECTION

Blood Flow, Blood Pressure, and Resistance

Blood flow, blood pressure, and resistance.

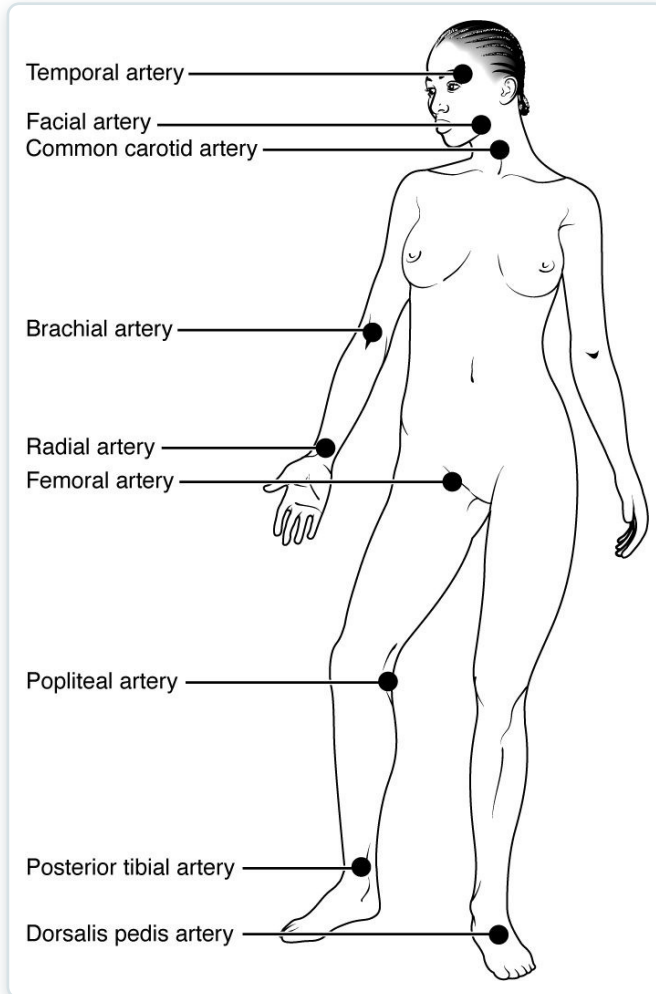
Blood Pressure Across the System

- Pressure is highest in the arteries
- Drops sharply in the arterioles
- Lowest in the veins
- The gradient drives blood flow



Blood pressure throughout the circulation

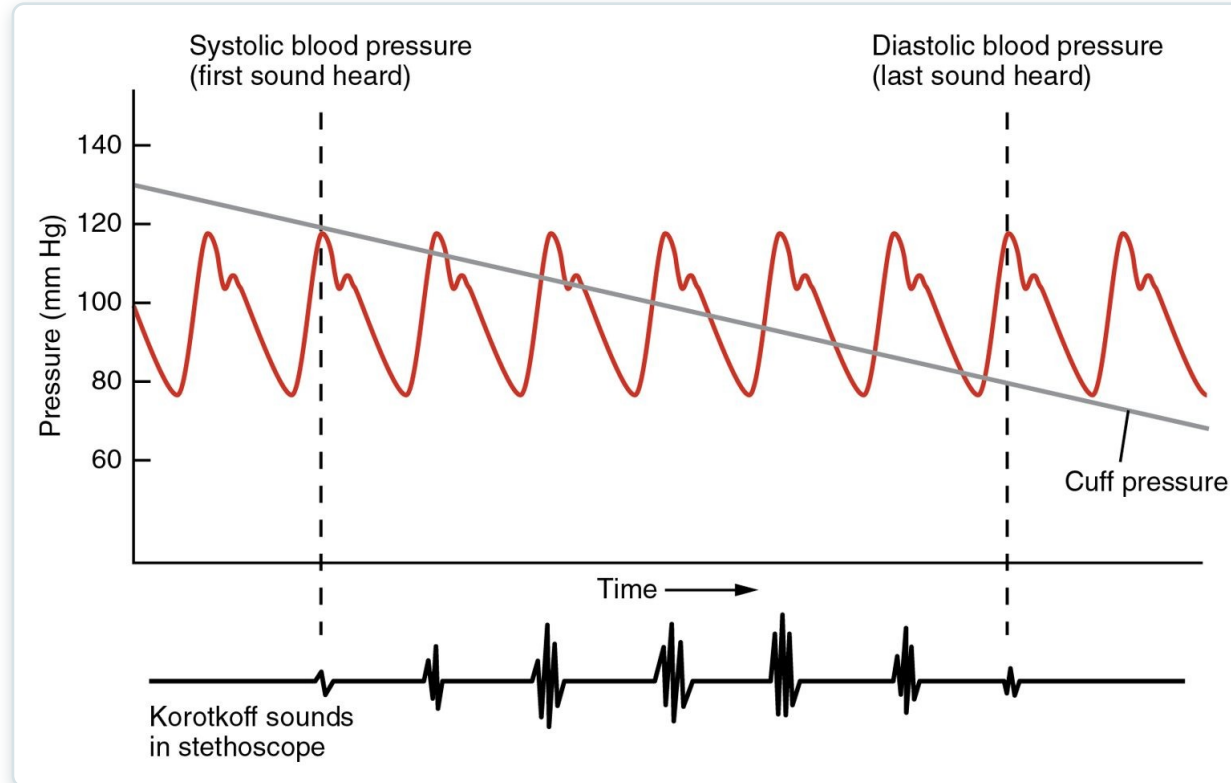
Pulse Points



Pulse points of the body

- Pulse = artery expanding with each beat
- Felt where arteries are near the surface
- Common sites: radial, carotid, femoral
- Used to assess rate and rhythm

Measuring Blood Pressure

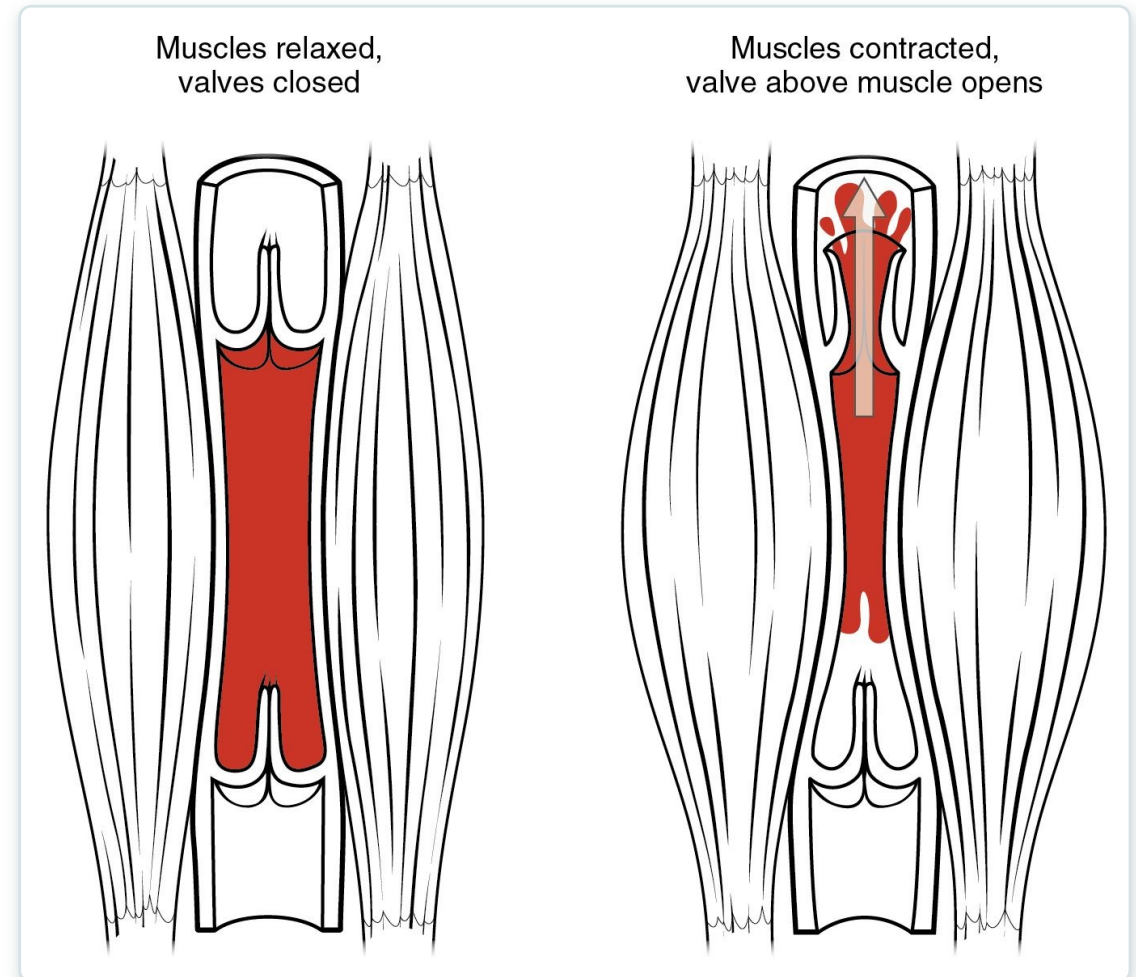


Measuring arterial blood pressure

- Measured with a cuff (sphygmomanometer)
- Systolic over diastolic
- Normal about 120/80 mmHg
- A key vital sign

The Skeletal Muscle Pump

- Veins need help returning blood
- Contracting muscles squeeze veins
- Valves keep blood moving up
- Aids venous return to the heart



The skeletal muscle pump



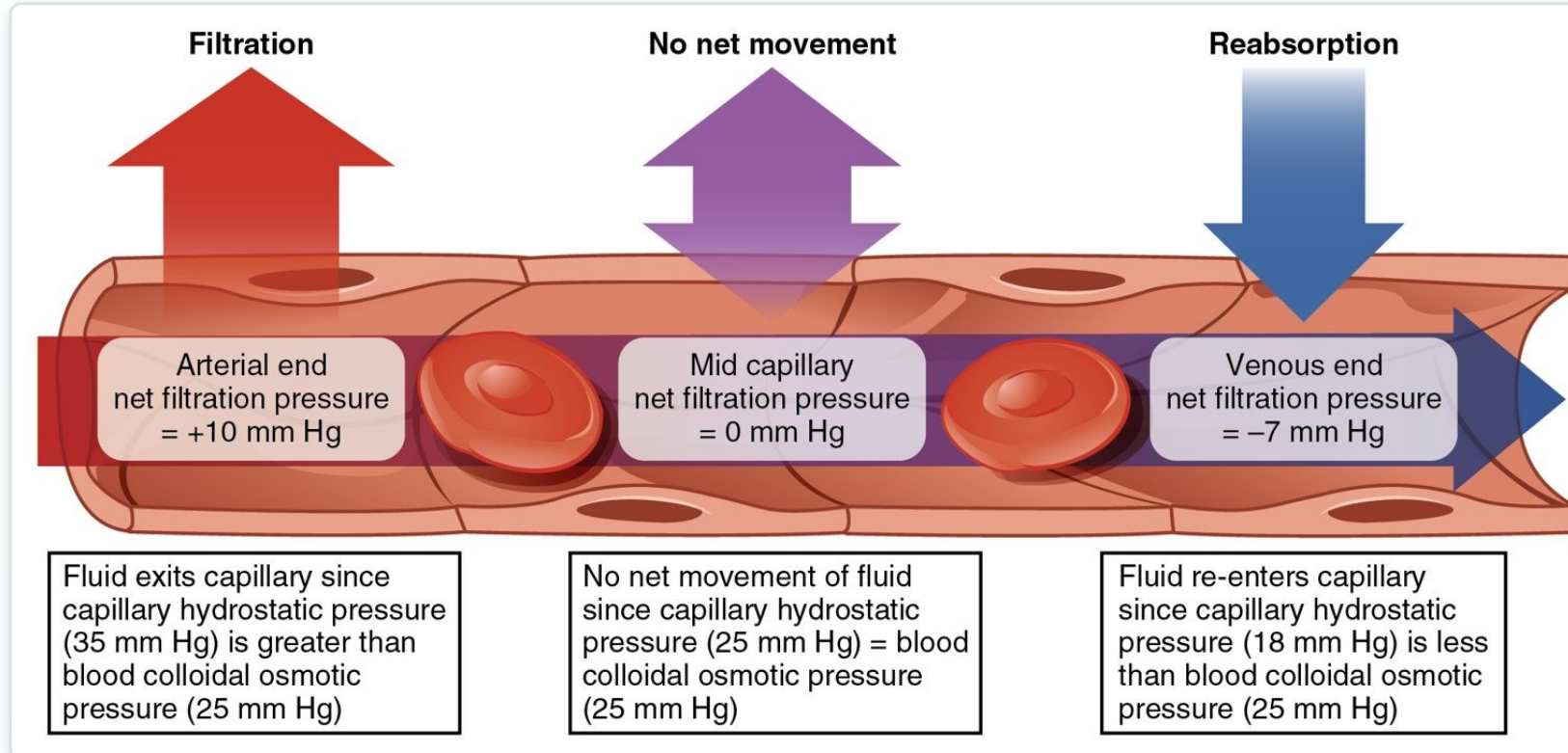
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SECTION

Capillary Exchange

Capillary exchange — where blood serves the tissues.

Capillary Exchange



Fluid exchange at the capillary

- Materials cross the thin capillary wall
- Pressure pushes fluid out (filtration)
- Osmosis pulls fluid back (reabsorption)
- Delivers nutrients, removes wastes



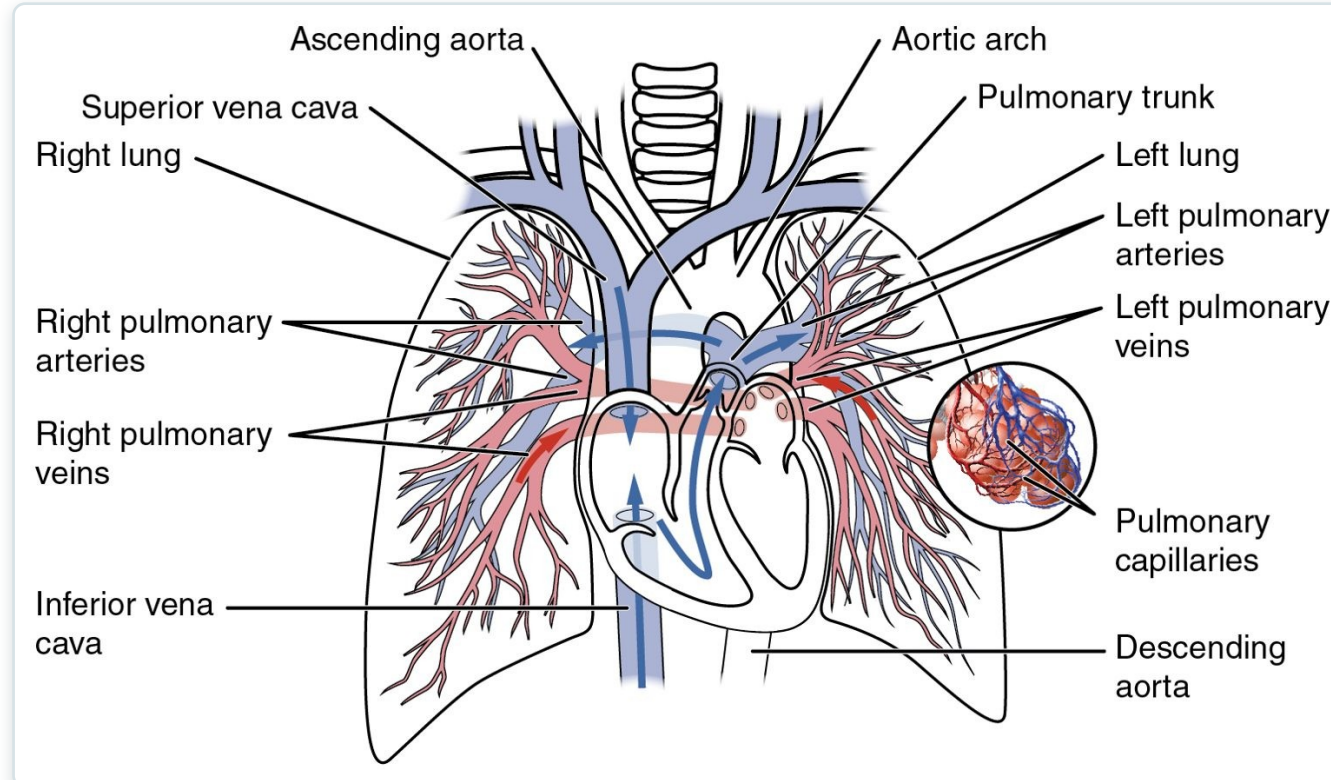
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SECTION

Circulatory Pathways

The major arteries, veins, and circulatory pathways.

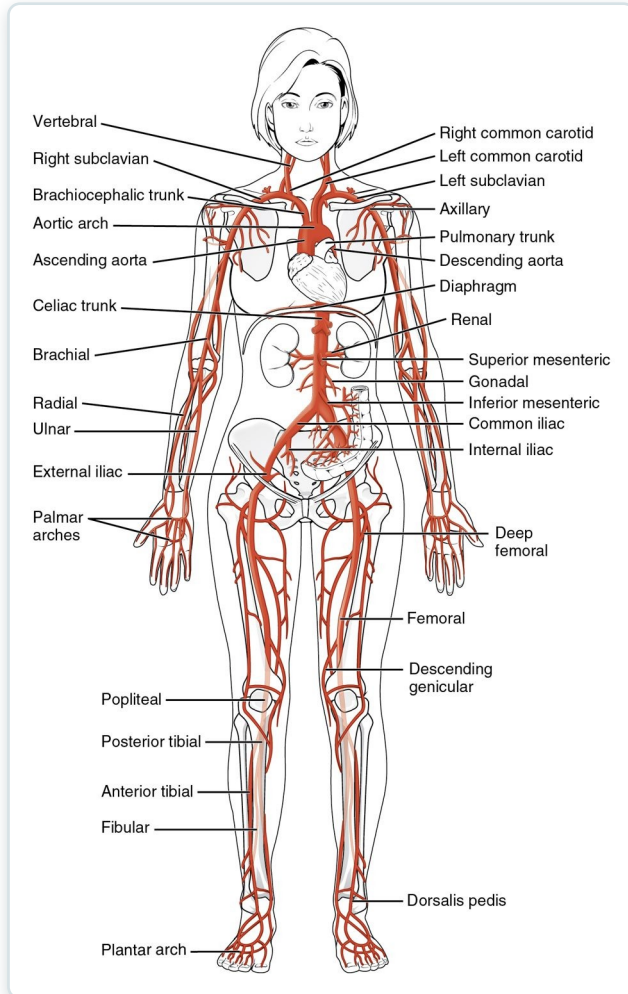
Pulmonary & Systemic Circuits



The pulmonary and systemic circuits

- Two linked circuits
- Pulmonary: heart ↔ lungs
- Systemic: heart ↔ body
- Run in series through the heart

Major Arteries



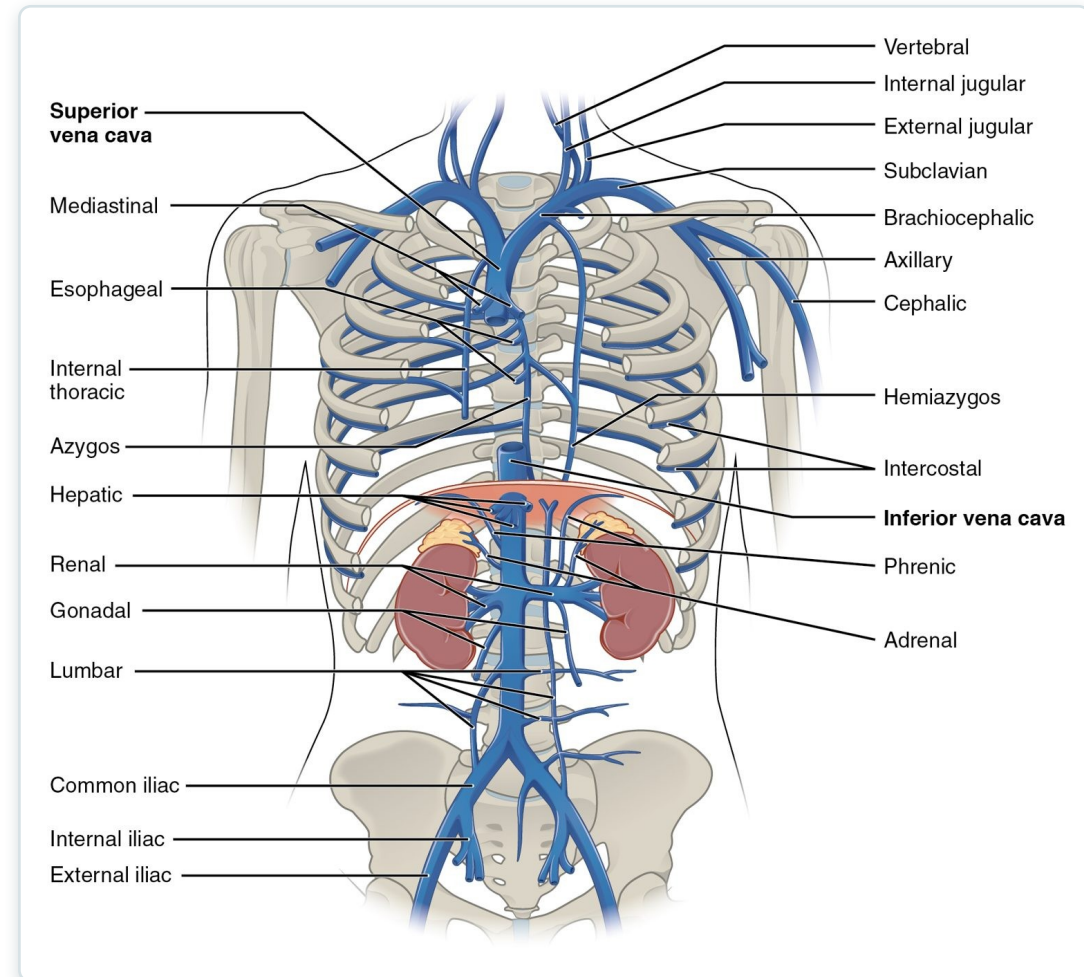
Major arteries of the body

- Arteries carry blood from the heart
- The aorta is the largest
- Branches supply every region
- Usually carry oxygen-rich blood

SECTION 20.5

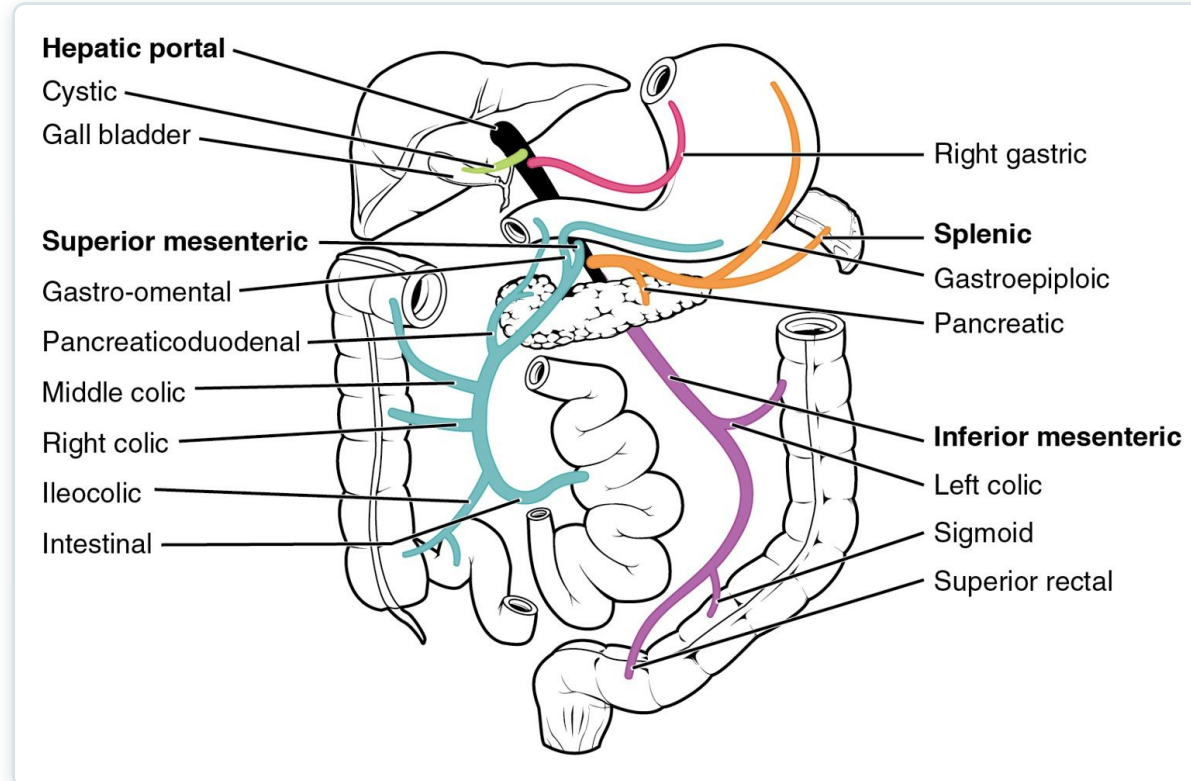
Major Veins

- Veins return blood to the heart
- The venae cavae are the largest
- Drain every body region
- Usually carry oxygen-poor blood



Major veins of the body

The Hepatic Portal System



The hepatic portal system

- A special detour through the liver
- Drains the digestive organs
- Carries nutrient-rich blood to the liver
- Liver processes it before circulation



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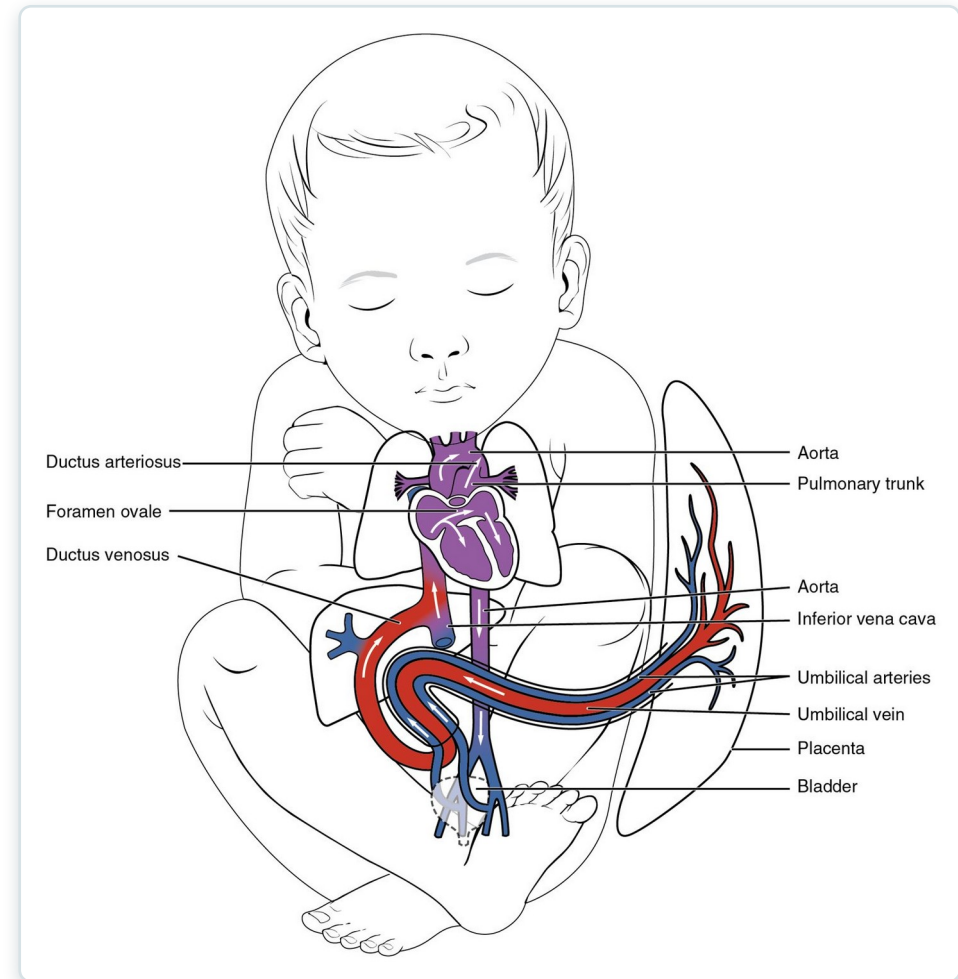
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Development of Blood Vessels and Fetal Circulation

Fetal circulation before birth.

Fetal Circulation

- The fetus gets oxygen from the placenta
- The lungs are bypassed before birth
- Shortcuts: ductus arteriosus, foramen ovale
- They close at birth



Fetal circulation

Key Takeaways

- Arteries carry blood from the heart, veins return it, and capillaries are where exchange occurs.
- Blood pressure is highest in the arteries and drives flow through the circuit.
- Capillary exchange delivers nutrients and removes wastes; imbalance causes edema.
- The systemic and pulmonary circuits carry blood to the body and the lungs.
- Fetal circulation bypasses the lungs through shunts that close at birth.

CLINICAL CONNECTIONS

Why Blood Vessels Matter in Practical Nursing

Hypertension

Chronically high blood pressure damages vessels and organs.

Atherosclerosis

Arterial plaque narrows vessels and raises clot risk.

Pulse & BP

Core vital signs reflecting cardiovascular status.

Edema

Fluid leaking from capillaries causes tissue swelling.

Varicose Veins

Failed venous valves let blood pool in the legs.

IV Access

Superficial veins are used for fluids, medications, and blood draws.